

# Paper Title

Author Name

*Abstract—*

*Index Terms—*

## I. INTRODUCTION

Physics-based simulation of radar returns from the ocean surface is important for the development and evaluation of maritime sensing algorithms, particularly when measured sea-clutter data are costly to collect and cannot cover the full range of wind, wave, and radar conditions. High-resolution observations have shown that sea clutter is strongly dependent on grazing angle and resolution, and that its amplitude statistics can depart substantially from the Rayleigh model and exhibit pronounced heavy tails [?], [?], [?]. A major source of these nonstationary returns is the sea spike, which may resemble a target response and therefore increase the false-alarm rate of maritime detection systems [?], [?]. Experimental and numerical studies have further associated intense sea spikes with steep or breaking wave crests, especially at low grazing angles [?], [?], [?]. These observations motivate sea-surface models that preserve the statistical properties of the background sea while explicitly representing the localized geometries responsible for anomalous scattering.

Linear random surfaces synthesized from directional wave spectra provide an efficient description of the multiscale energy distribution of wind-driven seas. In particular, the unified spectrum of Elfouhaily *et al.* has been widely used to generate realizations containing both gravity and gravity-capillary components [?]. However, a linear spectral realization remains approximately Gaussian and cannot adequately reproduce crest-trough asymmetry, crest sharpening, or the locally overturning geometry of a plunging breaker. Nonlinear transformations, including Creamer- and Lie-type formulations, improve the higher-order morphology of a spectral sea surface, but the resulting surface is generally still represented as a single-valued height field. Moreover, the short-wave components embedded in a broadband random realization do not necessarily form the persistent, directionally organized wave ridges that precede localized breaking. Consequently, a direct transition is still missing between a statistically valid nonlinear background sea and an explicit, spatially localized curling-wave geometry.

The electromagnetic consequences of breaking waves have been investigated extensively. Full-wave and hybrid numerical studies have explained the contributions of crest diffraction, rough-face scattering, and single- and multiple-bounce propagation to low-grazing-angle sea return [?], [?], [?], [?], [?], [?]. Facet-based models have subsequently embedded breaking-wave contributions into larger sea surfaces and reproduced changes in normalized radar cross section (NRCS), sea-spike behavior, and synthetic aperture radar imagery [?]. More recently, three-dimensional plunging-breaker models combined

with ray tracing have demonstrated that a breaker crest can generate a target-like peak in radar cross section and high-resolution range profiles [?]. These studies establish the importance of breaker geometry, but they mainly focus on electromagnetic scattering from prescribed or numerically generated breaking profiles. Constructing a computationally efficient, large-area random sea that connects background slope statistics, directionally organized short waves, and localized curling crests remains comparatively less explored.

This letter develops a hierarchical sea-surface model for controllable radar-echo simulation. First, an Elfouhaily directional spectrum is combined with a wind-dependent nonlinear Lie transformation to construct a non-Gaussian background surface. Second, a selected short-wave band is replaced by an equal-energy directional component, thereby introducing wind-driven ridges with a prescribed dominant propagation direction without duplicating spectral energy. Third, candidate crests are detected from their elevation, propagation-direction slope, and curvature, after which a compactly supported coordinate transformation is applied to form a localized forward-extending and downward-curling breaker. The resulting representation retains the large-area statistics of the background sea while allowing localized facets to develop the steep and overturning orientations required for radar-oriented analysis.

The model is evaluated at three complementary levels. The background and short-wave-enriched surfaces are first examined through spectral consistency, significant wave height, and along-wind and crosswind mean-square slopes, with Cox-Munk-type slope relations used as empirical references. The localized breaker is then assessed using crest geometry, facet-orientation statistics, radar-facing projected area, and paired unit-footprint scattering enhancement. Finally, the complete surface is coupled to a facet-based radar-return model, and the simulated range-time intensity patterns, temporal correlation, and amplitude probability distributions are compared with measured-clutter behavior and established mean-backscatter and clutter-statistics references [?], [?], [?], [?], [?], [?]. The main contributions are therefore: 1) an energy-preserving connection between a nonlinear spectral background and directionally organized short waves; 2) a crest-conditioned local transformation that introduces explicit curling geometry at low computational cost; and 3) a multilevel validation framework linking global sea-surface statistics, local breaker geometry and scattering, and dynamic radar observables.

## REFERENCES